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PV power forecasting based on data-driven models: a review

Priya Gupta and Rhythm Singh

Department of Hydro and Renewable Energy, Indian Institute of Technology Roorkee, Roorkee, India

ABSTRACT

Accurate PV power forecasting techniques are a prerequisite for the optimal management of the grid and its stability. This paper presents a review of the recent developments in the field of PV power forecasting, mainly focusing on the literature which uses ML techniques. The ML techniques (sub-branch of artificial intelligence) are extensively used due to their ability to solve nonlinear and complex data structures. PV power forecasting can either be direct, or indirect, which involves solar irradiance forecast model, plane of array irradiance estimation model, and PV performance model. This paper presents a review of both of these pathways of PV power forecasting based on the proposed methodology, forecast horizons and the considered input parameters. In case of unavailability of historical PV power for a new PV plant and in case of failure of real-time data acquisition, indirect PV power forecasting can be a viable alternative. Although the performance ranking of various ML models is complicated and no model is universal, recent studies suggest that methodologies like deep neural networks and ensemble or hybrid models outperform conventional methods for short-term PV forecasting. Recent articles also present the various intelligent optimisation and data-preparation techniques to improve performance accuracy.

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PV power forecasting; forecast horizon; data-driven models; machine learning; deep learning; ensemble methods; solar radiation forecasting; POA irradiance; PV performance models

1. Introduction

Conventional energy sources are losing importance because of global warming and the rapid depletion of fossil fuels. Non-conventional energy sources such as solar, wind, bio-mass, etc., are free from the above problems, and thus, are poised to play a significant role in the future. Peters et al. (Peters and Buonassisi 2000) have analysed the effect of global warming on the performance of PV installations in the future. Solar energy is becoming one of the most promising sources to supply power for residential, commercial and industrial loads (Ren, Suganthan, and Srikanth 2015). Nevertheless, one of the major problems with renewable energy sources is their unpredictable nature. PV plants generate different amounts of electricity based on the variation of solar radiation and other meteorological variables (Ismail et al. 2015). There can be significant variations in the power output not only over different days, but also from one hour to another, or even over few minutes (Kasu and Hara 2017). This has the potential to lead to load-generation mismatch in the grid, thus rendering PV power forecasting very important, particularly in grids with high penetration of PV (Espinar et al. 2013).

1.1. Challenges and the need to forecast solar PV power

The integration of renewable energy sources with the electrical grid has been gaining more importance and is also creating more challenges for electrical engineers and researchers. The intermittent and uncontrollable nature of solar energy increases the complexity of grid management and adds to the difficulty in balancing the generation and consumption of electrical energy (Lara-Fanego et al. 2012; Raza et al. 2016). A balance between generation and consumption has become a great challenge for electrical operators. Voltage fluctuations, low power quality, and low stability issues are some other problems that arise due to the uncontrollable nature of solar energy. Variability, uncertainty and asynchronous operations are the main technical challenges electrical operators have to deal in integrating renewable energy sources with the power grid (Kroposki 2017). So, for optimal management of the electrical grid, accurate forecasting of solar PV power is required (Paulescu et al. 2013). Solar power forecasting is also required for scheduling, approximating the reserves, dealing generated electrical power, better operation of the power grid, reducing the cost of produced electrical energy and congestion management. As the penetration of solar PV in the grid increases, the prediction of solar power also becomes more critical due to the above-mentioned problems in the power system. Researchers also suggest using storage systems with renewable energy prediction to control electricity variation. Storage systems absorb excess power, dampen the fluctuations and maintain a continuous flow of electricity. This paper aims to discuss and compare different forecasting techniques to estimate the PV power output in two different ways, i.e. (i) direct forecasting that predicts the power directly by using historical data of PV power and (ii) indirect forecasting, which uses solar irradiation forecasting, along with other meteorological variables that directly affect solar PV power production (Rana, Koprinska, and Agelidis 2016).

A number of techniques for PV power forecasting have been proposed based on physical assumptions of the atmosphere. NWP is the most commonly used physical model of forecasting. The variability in the atmosphere and its

CONTACT Rhythm Singh rhythmunleashed@gmail.com Department of Hydro and Renewable Energy, Indian Institute of Technology Roorkee, Roorkee, Uttarakhand 247667, India

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unpredictable nature make the numerical weather prediction model computationally severe and complex (Sheng et al. 2018). Due to the growth in the computer science field and its ability to deal with non-linearity, the popularity of machine learning techniques is increasing. Apart from direct and indirect PV power forecasting using ML techniques, this paper also reviews, classifies and compares various plane of array irradiance estimation models and PV performance models in the field of PV power forecasting.

This paper will help researchers working in the field of solar energy by providing useful information on PV power forecasting. The sections of this paper are organised as follows- Section 2 presents an overview of PV power forecasting methodologies, Section 3 presents a detailed literature review in the field of solar power forecasting, Section 4 presents the development of literature on PV power forecasting, and finally, Section 5 concludes the complete work.

2. Available PV power forecasting methodologies

Solar power forecasting can be divided into two groups- (1) Direct forecasting, that directly forecasts PV power. (2) Indirect forecasting, that is based on solar radiation forecasting. Several researchers have been working in this domain over the last few years. However, still, the accuracy of short-term forecasting is a challenging task for the researchers. Solar radiation/PV power forecasting is a non-linear problem, which depends upon several weather parameters. For a non-linear system, finding the proper parameter estimation method is a difficult task. Several methods have been proposed for forecasting in the literature; two major categories are (i) physical models and (ii) data-driven models. Choice of forecasting model depends upon the forecast horizon and the selected location. Numerical weather prediction is the most commonly used physical model. It uses a dynamic atmosphere model (to use cloud motion and direction) and mathematical equations to predict radiation for N-step ahead forecast horizon. Data-driven models are another category of forecasting. These models are based on the extraction of useful information from the input training data, and based on this information, they predict the output. The performance of these methods is susceptible to the quality of the training data. After a lot of research in this field, it has been concluded that machine learning-based models show better performance for short-term forecast horizons (readers can refer to the references mentioned in Section 3 of this article). This section presents a detailed overview of data-driven models, plane of array irradiance estimation models, and PV performance models in the field of PV power forecasting. (Figure 1) shows the block diagram of PV power forecasting.

2.1. Data-driven models

As the name suggests, data-driven models use historical data. Models are trained, or extract some useful information from the dataset, to predict the output. Length and quality of historical data are two main factors that affect the accuracy of forecasting. Time series and ML models are the two main categories of data-driven models. (Figure 2) represents the classification diagram.

2.1.1. Time-series models

Time series is a part of statistics. Four researchers Slutsky, Walker, Yaglom and Yule, first composed the idea of time series models such as auto regressive (AR), integrated (I) and moving average (MA) (Brockwell and Davis 2006). The combination of these three concepts produces ARMA and ARIMA models. These are the simplest models in the field of times series forecasting. These models use historical data points of the dependent variables, known as endogenous forecasting. Exogenous forecasting utilises a historical dataset of independent variables other than the dependent variable. For

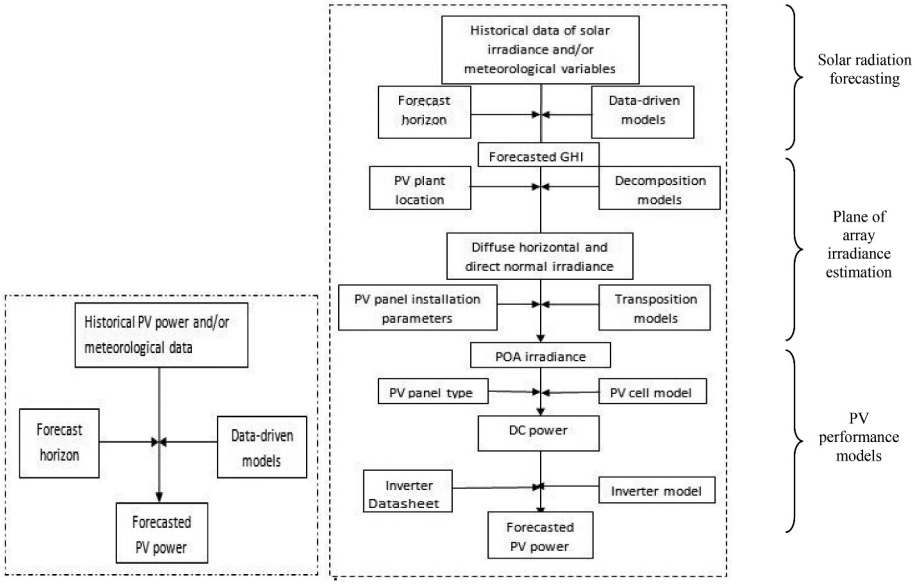


Figure 1. Block diagram of PV power forecasting: (a) direct forecasting and (b) indirect forecasting.

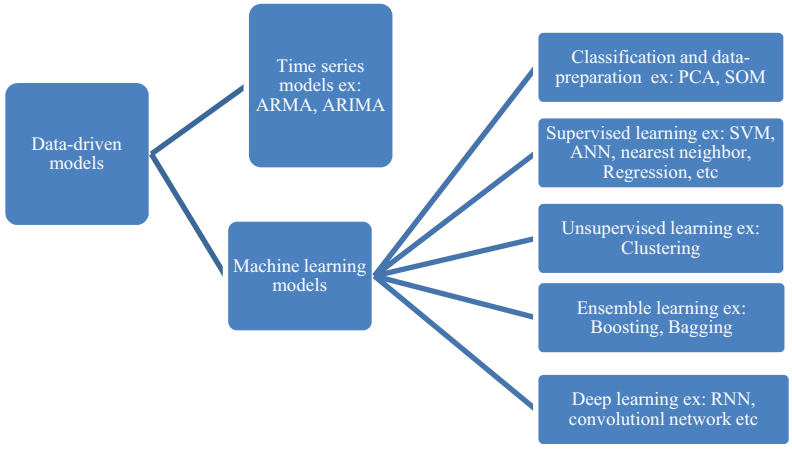


Figure 2. Classification of data-driven models.

exogenous time series forecasting, these models are termed as ARMAX and ARIMAX. There are some limitations associated with the traditional ARMA and ARIMA models:

(1) Missing values in the data set highly affect the accuracy of these models.
(2) Solar radiation and power forecasting is a non-linear problem, but these models cannot effectively deal non-linearity of the problems (Voyant et al. 2017b).
(3) These models are usually applied for univariate forecasting applications. But many problems in the real world depend upon more than one input variable.
(4) These models usually don't work well for the multi-step forecast horizon.

2.1.2. Machine learning models

ML is a sub-branch of computer science and also a sub-field of artificial intelligence techniques. Machine learning utilises historical data points to establish a relation between the input and output, even if the relationship between them is complex. So it is recommended to use proper data or prepare them to solve the problem effectively. Machine learning comprises a gamut of techniques; some of these have been discussed in the remainder of this section as follows.

Data-preparation and classification techniques: For the pre-processing of input data, data preparation and classification techniques are utilised with the forecasting model. Some of them are described as follows-

Principal component analysis (PCA): It is a linear transformation technique used for dimensionality reduction and feature extraction. This method transforms correlated variables into mutually uncorrelated variables using some orthogonal transformation. The principal components obtained from the Eigen vector of the covariance matrix can be lower or equal to the original variables. Most of the variance is achieved by the first principal components that indicate a high correlation between input variables (Alskaif et al. 2019).

Self-Organising map (SOM): Self-organising map is a feature extraction technique based on unsupervised learning. The technique is sometimes termed Kohonen map because it was first developed by professor Kohonen (Yang et al. 2014). It is a kind of ANN technique used to convert the input vectors in low dimensional representation and is therefore termed dimensionality reduction technique.

Wavelet transform (WT): It is a potent tool to decompose a time series in both time and frequency domain. It is handy for extracting some hidden local information from the time-series data. It is more powerful than the Fourier transform. It is not an ML technique, but applying wavelet transform with various ML models, including SVM, ANN (can deal with non-linearity of series) in the classification phase, improves the forecasting results effectively (Eseye, Zhang, and Zheng 2018; Wen et al. 2019).

Supervised learning: Supervised learning-based techniques learn from the training data set, which consists of some inputs and their desired outputs, to establish a general mapping rule for finding a relation between the inputs and the outputs. Some supervised learning-based ML techniques are as follows.

Regression: Regression analysis determines the functional relationship between variables and forecasting. When only one parameter is required, then the regression will be univariate, and in case two or more variables are involved, regression will be termed as multivariate. The type of regression model depends on the complexity or relationship between dependent and independent variables and the number of independent variables. Two common types of regression techniques are SLR and MLR (Sobri, Koohi-kamali, and Rahim 2020).

Artificial neural network (ANN): Neural network is an AI technique that resembles the brain in two aspects. It can deal with both linear and non-linear regression and classification problems. NN is usually a combination of three layers (i.e. input, hidden and output layer), activation function and summation node.

$$y = y(x; w) = a_0 + \sum_{j=1}^{m} (w_j f) \sum_{i=1}^{d} w_{ji} x_i \qquad (1)$$

Equation (1) defines a standard mapping function for input x and output y with d inputs and m hidden layers. MLP is the most commonly used structure of NN that is formed by an input layer, one or more hidden layers, and one output layer. Care must be taken while selecting the NN structure, as a too complex structure of NN can easily overfit the training data

(Voyant et al. 2017b). The complexity of the neural network depends upon the number of hidden layers and the dimension of weights (w). Network topology depends on the connection of neurons, and for better functioning, it is an important factor. Researchers are using hybrid methods for selecting NN topology that allows the neural network to achieve more gain and perform at its maximum capacity.

Support vector machine (SVM): SVM is a kernel-based ML technique used in both classification and regression tasks. The idea of a support vector machine was first introduced by Vapnik (Cortes and Vapnik 1995). These models require some kernels, can make fast computations, avoid overfitting of training data, and have good convergence. SVM models are simpler to implement and faster in computation compared to neural networks. These are the reasons that SVM models are preferred for time series forecasting problems. Based on the type of kernel, SVM models are classified into L-SVM and K-SVM. L-SVM models use linear kernel functions, whereas K-SVM models use a non-linear kernel function and can transform the data into higher dimensional space. Prediction calculated by SVM can be given by:

$$y = \sum_{i=1}^{n} \left(a_i k_{rbf}(x_i, x*) + \mathrm{b}\right) \tag{2}$$

Equation (2) represents the mathematical representation of predicted output value for the test case input $x*$, b is the bias parameter, a_i is the coefficient and x_i is the training set input vector at i^{th} instance with total n training vectors.

And the commonly used radial basis function kernel is given by:

$$k_{rbf}(x_p, x_q) = \exp\left[\frac{-(x_p - x_q)}{2\sigma^2}\right] \tag{3}$$

The radial basis function is the most commonly used kernel in the support vector machine regression and classification problems (Fan et al. 2018). Equation (3) represents a general mathematical form of this type of kernel function. The $(x_p - x_q)$ represents the relation between two feature vectors in terms of the squared Euclidean distance, and σ is the free parameter. The Gaussian function, polynomial kernel and sigmoid kernel are some other kernel functions.

Nearest neighbour: The k-NN neural network is one of the simplest types of ML algorithm that is also used for both classification and regression purposes. The k-NN algorithm assigns weights to all the neighbours. The weight of each neighbour is inversely proportional to the distance of the neighbour to its centroid (Pedro and Coimbra 2015). It means the weight assigned to the minimum distance neighbour will be the highest.

Unsupervised learning: Unsupervised learning does not need expert involvement to extract useful information from the input data, unlike supervised learning. This is why unsupervised learning algorithms cannot be used directly for regression and classification problems. Unsupervised learning-based models can find the hidden structure or useful information from the input data without knowing the output. Clustering is one of the crucial techniques of unsupervised learning to preprocess the data in the form of clusters of similar information. Some clustering techniques are as follows-

K-means clustering: Clustering is one of the most commonly used data analysis techniques to find the subgroup in the data points to form clusters. The data points in each cluster are of similar nature. K-means clustering is an unsupervised ML technique. The main aim of this technique is to extract useful information and relations between inputs in the form of clusters. These will then be useful to predict the output in the forecasting problems.

Hierarchical clustering: This is also a type of unsupervised technique that builds the hierarchy of clusters in a tree structure called 'dendrogram'. A dendrogram is a tree structure consisting of roots and leaves (Voyant et al. 2017b). Roots contain all the information of the clusters, and leaves correspond to individual information.

Ensemble learning: Ensemble learning is a hybrid ML technique that uses weak learners to predict the output and then combines these outputs to obtain a final prediction. Nowadays, researchers prefer ensemble models because they do not generate output by the estimation of only one model; instead, it combines the output of many weak learners. The reason that researchers are focusing on the use of ensemble models is that these models improve the stability and accuracy of single-stage machine learning algorithms. Depending on how the weak learners are connected, two types of ensemble learning models are formulated: Boosting and Bagging.

Boosting: Boosting is also an ML technique used in both classification and regression tasks. Boosting takes the advantage of some weak learners known as decision trees to predict the output. Boosting predicts the output in a stage-wise manner, or in a serial form by optimising some loss function (Cheng 2016). Boosting combines many weak learners and gives more preference to a bad classifier to form a strong learner. Gradient boosting is one of the examples of boosting technique.

Bagging: Bagging is a machine learning algorithm that is also gaining importance in the field of forecasting. The basic idea behind bagging is to generate multiple versions of decision trees or weak learners and get an aggregated strong predictor by combining the output of each decision tree. Weak learners can be formed either by taking one ML model with different sets of training data or by taking different ML forecasting models. Bagging is a unique form of averaging forecasting technique (Bbeiman 1996). Random forest is one of the examples of bagging techniques.

Deep learning (DL): Deep learning is a sub-class of machine learning and can be supervised or unsupervised. Due to the ability of artificial neural networks to deal with the non-linearity of the problem, researchers are focusing on the further development of ANN methods. Deep learning is a more advanced form of neural network with a memory unit and can apply in many fields, including forecasting. Some models based on supervised deep learning like recurrent neural networks (Guan, Zhang, and Yan 2019), deep belief networks, convolutional neural networks have started to produce excellent results in the field of solar energy forecasting. Compared to other forecasting methods like NWP physical model, persistence and other statistical methods, DL-based models can effectively extract deep features and information from the time series.

2.2. Plane of array irradiance estimation models

The plane of array irradiance estimation model comprises two empirical models, i.e. decomposition and transposition models. These two models are termed as combination models. (Figure 3) shows the structure diagram to the model plane of array irradiance using modelled GHI which can be obtained by utilising various data-driven and physical models.

2.2.1. Decomposition models

Decomposition models are used to decompose the GHI data into direct normal and diffuse horizontal irradiance. Two substantial physical quantities in this context are clearness index(Kt) and a diffuse fraction(DF) (Lave et al. 2015). The accuracy of decomposition models depends upon the relationship of the clearness index and the diffuse fraction. These models are empirical because their equations are not formally acquired from physical rules but involve some coefficients. These coefficients are derived by collecting solar irradiance data from various locations.

2.2.2. Transposition models

Transposition models transpose direct normal and diffuse horizontal irradiance into global solar irradiance on a tilted plane. Plane of array (POA) irradiance(Et) is divided into the following components: **(1)** direct normal component of POA (Etb), **(2)** sky diffuse component of POA (Etd) and **(3)** reflected ground-diffuse component of POA (Etg). The equation of POA can be represented as in Equation (4).

$$E_t = E_{tb} + E_{td} + E_{tg} \tag{4}$$

The direct irradiance incident on POA can be calculated directly by direct normal irradiance, as given in Equation (5).

$$E_{tb} = DNI * \cos(AOI) \tag{5}$$

Where AOI is the angle of incidence of solar radiation on POA and DNI is direct normal irradiance on a horizontal plane.

The only difference in various transposition models lies in the sky-diffuse component. The sky-diffuse component of POA irradiance depends on the amount of sky seen from the surface (Cui et al. 2019).

The reflected ground diffuse POA irradiance component is a function of GHI, ground albedo (ρ) and surface tilt from horizontal (β), as given in Equation (6).

$$E_{tg} = GHI * \rho * \frac{1 - \cos\beta}{2} \tag{6}$$

2.3. PV performance models

2.3.1. PV cell model

To establish PV power forecasting, PV cell model is the core component. The voltage-current equation of the single diode PV cell model (Huang et al. 2010) can be represented by Equation (7).

$$I = I_p - I_0\left\{\exp\left[\frac{q(V + IR_s)}{nkT}\right] - 1\right\} - \frac{(V + IR_s)}{R_{sh}} \tag{7}$$

Where Ip is the current generated from light, Io is the reverse saturation current, k is the Boltzmann constant, n is the ideality factor, q is the charge, Rs is the series resistance, Rsh is the shunt resistance, and T is the absolute temperature. The single diode model is easy to implement. For a more accurate forecast, researchers also suggest the use of the two diode model. The two diode model is comparatively complex but can provide more precise results. Equation (8) represents the current-voltage equation of the two diode models (Bouralou et al. 2015).

$$I = I_p - I_{01}\left\{\exp\left[\frac{q(V + IR_s)}{kT}\right] - 1\right\} - I_{02}\left\{\exp\left[\frac{q(V + IR_s)}{2kT}\right] - 1\right\} - \frac{(V + IR_s)}{R_{sh}} \tag{8}$$

The equivalent circuit of the two diode model is the same as the one diode model with an additional parallel-connected diode. I_{01} and I_{02} are the reverse saturation current in the first and the second diodes, respectively.

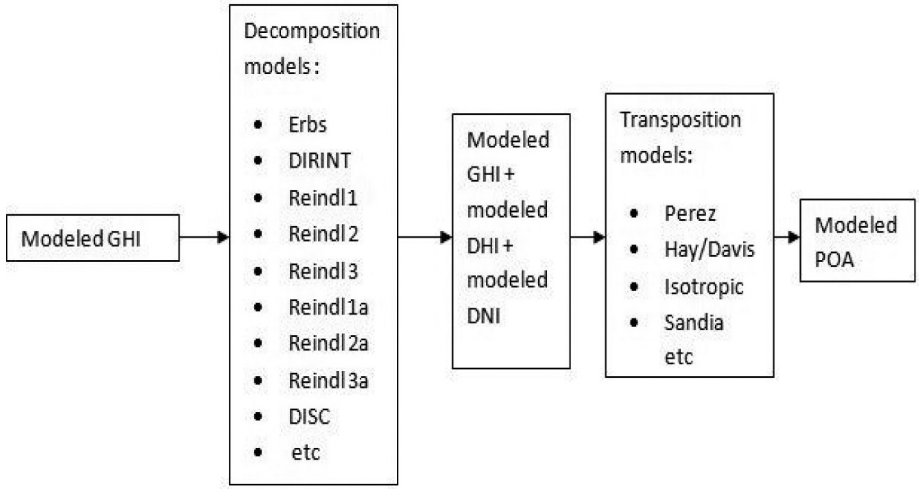


Figure 3. Structure to model plane of array irradiance.

2.3.2. Inverter model

PV array and inverters are the main components of the grid-connected power system. Therefore, accurate simulation of an inverter is the critical deciding factor for accurate forecasting. Few inverters can achieve the efficiency specified on the inverter datasheet because of the loss associated with the inverter components. Sandia National Laboratory proposed an inverter which considers the loss associated with the inverter components and also shows the high accuracy (King, Gonzalez, and Galbraith 2007). Authors in (Cui et al. 2019; Liaoliao et al. 2017) consider this inverter model for the application of PV power forecasting.

3. Literature review

In this section, we have analysed the state-of-the-art of literature on solar PV power forecasting. The literature review has been organised into two groups, i.e. direct forecasting and indirect forecasting.

3.1. Direct forecasting

As discussed previously that ML techniques need some historical data for the training of the model, so, for direct PV power forecasting, the availability of such historical PV power data is an essential requirement. This can be possible if the PV plant is old and equipped with data loggers and a SCADA system. Papers based on data-driven models classification in Section 2 for the direct PV power forecasting, are categorised as follows.

Time series models – (Pedro and Coimbra 2012), (Liu, Zhan, and Bai 2019), (Yin et al. 2020) **Classification and data-preparation** – (Yang et al. 2014), (Eseye, Zhang, and Zheng 2018), (Wen et al. 2019), (Chen et al. 2011), (Haque, Nehrir, and Mandal 2013), (Zhu et al. 2016), (Chang and Lu 2020), (P.Li et al. 2020)**, Supervised learning (1) NN**- (Wen et al. 2019), (Chen et al. 2011), (Cococcioni, D'Andrea, and Lazzerini 2011), (Pedro and Coimbra 2012), (J. Liu et al. 2015), (Ramsami and Oree 2015), (Zhu et al. 2016), (Hossain et al. 2017), (J.Wang, Ran, and Zhou 2017), (Gigoni et al. 2018), (Raza, Mithulananthan, and Summerfield 2018), (Rosiek, Alonso-Montesinos, and Batlles 2018), (Yadav et al. 2018), (Visser, Alskaif, and Van Sark 2019), (Sharadga, Hajimirza, and Balog 2020), (Yin et al. 2020), (Dewangan, Singh, and Chakrabarti 2020), (Zhou et al. 2020), (Z.F. Liu et al. 2020) **(2) SVM**- (Rana, Koprinska, and Agelidis 2016), (Yang et al. 2014), (Eseye, Zhang, and Zheng 2018), (Shi et al. 2012), (Gigoni et al. 2018), (L.Liu, Zhan, and Bai 2019), (Visser, Alskaif, and Van Sark 2019), (VanDeventer et al. 2019), (Dewangan, Singh, and Chakrabarti 2020), (Pan et al. 2020) **(3) Regression** – (Sheng et al. 2018), (Almeida, Perpiñán, and Narvarte 2015), (Ramsami and Oree 2015), (Wang, Su, and Shu 2016), (Gigoni et al. 2018), (L. Liu, Zhan, and Bai 2019), (Visser, Alskaif, and Van Sark 2019) **(4) Nearest neighbour** – (Pedro and Coimbra 2012), (Gigoni et al. 2018)**, Ensemble learning (1) Boosting** – (Visser, Alskaif, and Van Sark 2019) **(2) Bagging** – (Rana, Koprinska, and Agelidis 2016), (El-Baz, Tzscheutschler, and Wagner 2018), (Gigoni et al. 2018), (L.Liu, Zhan, and Bai 2019), (Visser, Alskaif, and Van Sark 2019), (Dewangan, Singh, and Chakrabarti 2020)**, Deep learning** – (Chang and Lu 2020), (P. Li et al. 2020), (Zheng et al. 2020), (Ali Jallal, Chabaa, and Zeroual 2020), (Yin et al. 2020), (Dewangan, Singh, and Chakrabarti 2020) A summary of the literature on direct PV power forecasting is given in (Table 1). Readers can refer to these references for more detail.

It can be seen from (Table 1) that artificial neural network (ANN), and support vector machine (SVM) are the two most often used machine learning algorithms in the field of PV forecasting. Several studies in the area of ANN validate the better non-linear fitting ability of neural networks compared to time series models. Broad research in neural networks, from an early simple architecture to a late deep configuration, results from the better performance of these networks. Sharadga, Hajimirza, and Balog (2020) recently presented a new deep learning model Bi-LSTM (Bi-directional LSTM), for PV power forecasting. After comparing the results of various structures of neural networks (NNs), and time-series models (ARMA, ARIMA, and SARIMA), the prediction accuracy of NNs was reported higher with less computational time. Unlike recurrent neural network (RNN), LSTM contains a memory unit that helps to keep the long spans data and can also solve the gradient descent problem. That's why LSTM is helpful to extract the temporal information from the time-series data. Similarly, a deep belief network (DBN) has been presented by Chang and Lu (2020) to learn the non-linear features from the previous PV power time-series data.

The core of machine learning algorithms is the selection of suitable hyperparameters values that impact forecasting accuracy. For example, C and gamma are the two important parameters of SVM, whose inappropriate values are responsible for the overfitting and underfitting issues. Presently, scholars are adopting intelligent optimisers for hyperparameter tuning of machine learning algorithms. Pan et al. (2020) reported the improvement in R^2 score (coefficient of determination) from 0.991 to 0.997 by utilising an improved ACO to optimise SVM parameters. Similarly, Zhou et al. (2020) adopted a genetic algorithm, and Liu et al. (2020) adopted an improved chicken swarm optimisation to tune the hyperparameters of extreme learning machine (ELM), and both reported a better forecast accuracy with the incorporation of optimisation algorithms. Further, Zheng et al. (2020) and Ali Jallal, Chabaa, and Zeroual (2020) presented deep learning-based models for PV power forecasting by optimising the parameters with PSO and RODDPSO techniques, respectively.

Besides this, researchers also combine different classification and data-preparation techniques before building any machine learning-based forecasting model to ensure the extraction of reliable information from the time-series data. As shown in (Table 1), Chang and Lu (2020) applied a grey theory on the raw data of PV power before building deep belief network to convert the irregular data into new data with strong regularity. Similarly, Li et al. (2020) involves applying wavelet packet decomposition to model the non-linearity of the original time-series data by decomposing the signal into several sub-series. While comparing the results of

Table 1. Summary of literature review on direct PV power forecasting.

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
(Chen et al. 2011) **China**	1-d	1 year	PV power, weather forecasted data (solar irradiance, air temp., wind speed and direction, cloud, sunshine hours, air pressure, and humidity	RBFNN, SOM, K-fold cross-validation	• The weather classification factor improves the performance of the neural network.
(Cococcioni, D'Andrea, and Lazzerini 2011) **Italy**	1-d	1 year	Solar irradiance and time as exogenous, PV power as endogenous	FFNN	• The proposed model obtains errors less than 5% for 1-d ahead horizon and can be extended for a broader forecast horizon.
(Pedro and Coimbra 2012) **USA**	1-h, 2-h	1 year 8 months	PV power	Persistent, ARIMA, K-NN, ANN, GA-ANN	• GA-ANN improves the performance by optimising the input variables.
(Shi et al. 2012) **China**	1-d	10 months	PV power, predicted daily temperature from the weather report	SVM model with weather classification	• The next day weather data report helps the SVM model to obtain good results.
(Haque, Nehrir, and Mandal 2013) **USA**	12-h	NA	Historical PV power, solar irradiance	WT+fuzzy ARTMAP (FA) +FF	• FF algorithm optimises the parameters of the FA model.
(Yang et al. 2014) **China**	1-d	1 year	Historical PV power, temperature, precipitation, solar irradiance	SOM+LVQ SVR+Fuzzy inference method	• SOM and LVQ classifiy the PV output power patterns into different weather types by extracting some features.
(Almeida, Perpiñán, and Narvarte 2015) **Spain**	1-d	1 year	AC PV power and meteorological variables	Non-parametric model, quantile regression forest	• Forecasted meteorological variables from NWP are utilised in a non-parametric PV model.
(J. Liu et al. 2015) **China**	24-h	NA	PV power, meteorological data (temperature, humidity, wind speed), Aerosol index	BP neural network	• The aerosol index reduces the prediction error under cloudy days.
(Ramsami and Oree 2015) **Mauritius**	24-h	11 months	PV Power, air temperature, pressure, humidity, rainfall, radiation, wind direction, and speed	Single-stage GRNN, MLR, FFNN and with SR	• ANN-based models gave a better performance than MLR schemes.
(Rana, Koprinska, and Agelidis 2016) **Australia**	5–60 min	NA	PV power, irradiance, wind data, temperature, humidity	Ensemble neural network, SVR, correlation-based variable selection	• A univariate model with correlation-based variable selection, ensemble neural network, and with non- iterative approach gives the best results for short-term forecasting horizon.
(Wang, Su, and Shu 2016) **China**	1-d	1 year 4 months	PV power, solar insolation, pressure	RPFLRM	• Regularised PFLRM is simpler and better than PFLRM.
(Zhu et al. 2016) **China**	1-d	NA	PV power	Wavelet decomposition and ANN hybrid model	• A hybrid model of WD and ANN predicts power with more accuracy and less computation time than the ANN model.
(Hossain et al. 2017) **Malaysia**	1-h, 1-d	1 year	PV power, Solar irradiance, wind speed, ambient temperature, module temperature	Extreme learning machine (ELM)	• ELM model is an efficient machine learning approach with high computational speed.
(Lave et al. 2017) **USA**	NA	1 year	Measured and forecasted data of PV power, load, net load	Experimental	• The authors identified spatial and temporal patterns in forecast errors to improve the accuracy of the various ML forecasting tools.
(J. Wang, Ran, and Zhou 2017) **China**	1-h	NA	PV power, Solar irradiance	ELM, OS-ELM, FOS-ELM (ELM with forgetting mechanism	• FOS-ELM algorithm shows higher accuracy followed by ELM and OS-ELM algorithm.
(El-Baz, Tzscheutschler, and Wagner 2018) **Germany**	1-d	1 year	PV power, forecasted weather data (temp., wind speed and direction, humidity, and cloudiness factor)	Clear sky model, bagging regression tree, probabilistic analysis	• The deviation of estimated PV power and the actual PV power measurements necessitates the need for probabilistic analysis.
(Eseye, Zhang, and Zheng 2018) **China**	1-d	1 year	SCADA PV power, NWP meteorological	Wavelet decomposition + SVM+PSO hybrid method	• The proposed method is suitable for short-term forecasting. It is found that as the forecast horizon increases, the accuracy of the model decreases.
(Gigoni et al. 2018) **Italy**	24-h	2 years 7 months	Global, diffuse, and direct radiation data, temperature data	GB (grey box model), QRF, K-NN, ANN, SVR, ensemble of models	• Ensemble models provide good results under all types of weather conditions.
(Raza, Mithulananthan, and Summerfield 2018) **Australia**	1-d	1 year	PV power, meteorological data (air temperature, humidity, wind speed, solar irradiance)	FNN, ELM, NewCF models LM, SCG, BP algorithms	• NewCF ensembles performed well as compared to FNN and ELM ensemble models.
(Rosiek, Alonso-Montesinos, and Batlles 2018) **Spain**	3-h	1 year	Global, direct, diffuse irradiance, wind speed and relative humidity, power output, air temperature, cloud images	Artificial neural networks (ANN) with satellite images	• The authors proposed a new methodology to forecast PV power using satellite-based forecasted global irradiance.
(Sheng et al. 2018) **Spain**	short term forecast	6 days	PV power, f photosynthetic active radiation, ambient temp., humidity, solar irradiance, precipitation and wind speed, and direction	Weighted gaussian process regression (WLOF)	• WLOF outperforms the other comparison models by extracting features from a high-dimensional dataset.
(Yadav et al. 2018) **India**	24-h	1 year	Solar radiation (SR), Back surface module temperature	RBFNN, Relief attribute evaluator algorithm	• SR found more relevant for predicting daily array yield with rank value 0.0077.

(*Continued*)

Table 1. (Continued).

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
(Seyedmahmoudian et al. 2018) **Australia**	1 h, 2 h, and 4 h	2100 hourly samples for training	PV power, solar irradiance, air temperature, relative humidity	DEPSO (hybrid of differential evolution and PSO)	• The proposed model outperformed the single-stage models (DE and PSO) for all considered forecast horizons.
(J. Huang et al. 2019) **Australia**	10 sec – 10 min	1 year	Time series of solar PV power, sky images	Statistical and sky cam based model	• A combination of statistical and sky cam-based approaches improves the forecasting accuracy compared to an individual approach.
(L. Liu, Zhan, and Bai 2019) **Australia**	1-d	5 years	Historical PV power, Wind speed, and direction, weather temperature, GHI, humidity, DHI	Ensemble of SVM, MLP, MARS	• The ensemble model performed better in both the training and forecasting stage compared to single-stage models.
(Visser, Alskaif, and Van Sark 2019) **Netherland**	1-d	3 years	Past PV power generation, weather variables (pressure, temperature, cloud cover, wind vectors, clear sky radiation, precipitation)	MLR, LASSO, L-SVM, K-SVM, GB, RF, FFNN	• GB and RF outperform both on individually and aggregation of 152 PV sites. • Aggregation of PV sites improves the result by reducing the variability effect.
(Wen et al. 2019) **Oregon**	1-h, 3-h, 6-h	1 year	PV power, solar irradiance, temperature	WT+RBFNN+PSO (point forecast) with bootstrap and QR as the probabilistic forecast	• Probabilistic analysis is helpful to determine the uncertainty in the results of point forecasts. • The Bootstrap probabilistic method outperforms the QR probabilistic method.
(VanDeventer et al. 2019) **Australia**	1-h	278 days for training	PV power, solar irradiance, air temperature	GASVM (genetic algorithm-based SVM)	• The involvement of GA with SVM has reduced the RMSE and MAPE by 669.624 W and 98.76%, respectively.
(Akhter et al. 2019) **Malaysia**	NA	NA	PV power and weather conditions	Metaheuristic and machine learning models (Review article)	• However, the study suggests that metaheuristic models combine different techniques and show improved forecast accuracy, but a tradeoff is also necessary between accuracy and other constraints (cost and computational complexity).
(Chang and Lu 2020) **Taiwan**	1-d	1 year	PV power	GT-DBN (grey theory-based deep belief neural network)	• The proposed model mitigates over-fitting problems by smoothing input data series.
(P. Li et al. 2020) **China**	1-h	2 years	PV power, GHI, DHI, ambient temperature, wind speed, relative humidity	WPD-LSTM	• The proposed model outperforms LSTM, GRU, RNN, and MLP models
(Zheng et al. 2020) **China**	30-min	1 year	PV power, sunshine, temperature, wind speed, humidity	PSO-LSTM	• PSO has been used to optimise the parameters of LSTM. The proposed model outperforms LSTM, ANN, and extreme gradient boosting.
(Ali Jallal, Chabaa, and Zeroual 2020) **Morocco**	1-h	2 years	PV energy, relative humidity, wind speed and direction, GHI, DHI, rainfall, temperature	DNN-RODDPSO	• The proposed model shows high performance with the integration of a selective layer and an automatic input selection hidden layer.
(Sharadga, Hajimirza, and Balog 2020) **USA**	1,2,3 h ahead	13/01/2010 to 29/10/2010	PV power	BI-LSTM, LSTM, FFNN, MLP, LRNN, ARMA, ARIMA, SARIMA	• Neural network shows more accuracy and less computation time compared to statistical models for time-series prediction.
(Yin et al. 2020) **China**	1-d	1 year	PV power, solar irradiance, wind speed, temperature, humidity	ELM with the non-iterative correction method	• The proposed non-iterative correction algorithm helps to make the real-time correction and to reduce the forecast error.
(Dewangan, Singh, and Chakrabarti 2020) **India**	1-d	2 years	NWP-derived variables like solar irradiance, cloud cover, precipitation, relative humidity, pressure	Persistence, linear regression, FFNN, LSTM, SVR, GPR, ensemble trees	• The authors proposed a combined forecast methodology based on mean, median, linear, and non-linear regression to determine the best model from the various similar level (performing) models.
(Zhou et al. 2020) **China**	1-d	January 2017-October 2018	Max., min and average ambient temperature, average GHI, and DHI	SDA-GA-ELM (similar day analysis-Genetic algorithm-Extreme learning machine)	• The proposed model outperformed the SVM, SDA-SVM, BPNN, persistence, SDA-BPNN, ELM and SDA-ELM models for the same dataset.
(Z.F. Liu et al. 2020) **China**	5 min	_	PV power, wind speed, radiation intensity, temperature	ICSO-ELM (Improved chicken swarm optimiser-Extreme learning machine)	• The proposed ICSO-ELM model is tested for three weather conditions, and the results were found better than other ML models, including CSO-ELM.
(Pan et al. 2020) **China**	5-min	2 years	PV power, temperature, relative humidity, GHI, DHI, sampling time, and wind direction	SVM-IACO (Improved ACO)	• The authors present some improvements in the traditional ACO algorithm to optimise the parameters of the SVM model.

classification and data-preparation techniques-based models with the single-stage models, it was reported that hybrid models perform better and show better performance accuracy compared to single-stage models. However, hybrid models may increase the cost and computational complexity. Therefore, a tradeoff between performance accuracy and other constraints is required while considering the hybrid models (Akhter et al. 2019).

3.2. Indirect forecasting

In the case of a newly developed PV plant, historical data of PV power of such plant may not be available for the training of ML models. In that case, direct forecasting from such models is not possible. However, due to the high correlation between solar irradiance and PV power, indirect forecasting can be implemented in such cases. The procedure to forecast PV power by this pathway is given as follows.

Step 1: Collect solar irradiance and meteorological data, PV plant location information, PV panel and inverter characteristics.

Step 2: Utilise data-driven models to forecast solar irradiance on a horizontal plane. (Researchers also use NWP models, but this paper aims to focus on ML techniques)

Step 3: Use combination models to calculate the plane of array solar irradiance.

Step 4: POA irradiance is applied as an input in PV performance models to forecast solar power.

The details are given in the following three subsections.

3.2.1. Solar radiation forecasting

Papers using data-driven models classified in Section 2, for the solar radiation forecasting on horizontal plane are categorised as follows: **Time series models –** (Fouilloy et al. 2018), (Marchesoni-Acland et al. 2019)**, Classification and data-preparation** – (Deo, Wen, and Qi 2016), (Shamshirband et al. 2016), (Bouzgou and Gueymard 2017), (S. Sun et al. 2018), (Lan et al. 2019), (Prasad et al. 2019), (Prasad et al. 2020), (Zambrano and Giraldo 2020), (Gupta and Singh 2021)**, Supervised learning (1) NN** –(Zeng and Qiao 2013), (Shamshirband et al. 2015), (Azimi, Ghayekhloo, and Ghofrani 2016), (Çelik, Teke, and Yildirim 2016), (Jiménez-Pérez and Mora-López 2016), (Aybar-Ruiz et al. 2016), (Gutierrez-Corea et al. 2016), (Wang et al. 2016), (Bouzgou and Gueymard 2017), (Mousavi, Mostafavi, and Jiao 2017), (Voyant et al. 2017a), (Alfadda, Rahman, and Pipattanasomporn 2018), (Fouilloy et al. 2018), (Khosravi et al. 2018), (Meenal and Selvakumar 2018), (Rao, Premalatha, and Naveen 2018), (Benali et al. 2019), (Cornejo-Bueno et al. 2019), (Lan et al. 2019), (Marzouq et al. 2019), (Yagli, Yang, and Srinivasan 2019), (Y. Feng et al. 2020), (Marzouq et al. 2020), (Pang, Niu, and O'Neill 2020), (Ben Ammar, Ben Ammar, and Oualha 2020), (Zambrano and Giraldo 2020), (H. Wang et al. 2020) **(2) SVM –** (Fan et al. 2018), (Zeng and Qiao 2013), (Belaid and Mellit 2016), (Deo, Wen, and Qi 2016), (Jang et al. 2016), (J. Li et al. 2016), (Jiménez-Pérez and Mora-López 2016), (Shamshirband et al. 2016), (Baser and Demirhan 2017), (Alfadda, Rahman, and Pipattanasomporn 2018), (Fouilloy et al. 2018), (Khosravi et al. 2018), (Meenal and Selvakumar 2018), (S. Sun et al. 2018), (Yao et al. 2018), (Cornejo-Bueno et al. 2019), (Yagli, Yang, and Srinivasan 2019), (Fan et al. 2020), (Prasad et al. 2020), (YagYagli et al. 2020), (Huertas-Tato et al. 2020), (Gupta and Singh 2021) **(3) Regression –** (Lou et al. 2016), **(4) Nearest neighbour** –(Alfadda, Rahman, and Pipattanasomporn 2018), (Ben Ammar, Ben Ammar, and Oualha 2020), **Clustering (Unsupervised learning) –** (Azimi, Ghayekhloo, and Ghofrani 2016), (Jiménez-Pérez and Mora-López 2016), (S. Sun et al. 2018), (Ben Ammar, Ben Ammar, and Oualha 2020)**, Ensemble learning (1) Boosting –** (Lou et al. 2016), (Hassan et al. 2017), (Fouilloy et al. 2018), (Pedro et al. 2018), (Yagli, Yang, and Srinivasan 2019), (Lee et al. 2020), (Kumari and Toshniwal 2021) **(2) Bagging –** (Jiménez-Pérez and Mora-López 2016), (H. Sun et al. 2016), (Hassan et al. 2017), (Ibrahim and Khatib 2017), (Voyant et al. 2017a), (Fouilloy et al. 2018), (Yagli, Yang, and Srinivasan 2019), (YagYagli et al. 2020), (Babar et al. 2020), (Narvaez et al. 2020), (Lee et al. 2020), (Gupta and Singh 2021) **Deep learning –** (Mishra and Palanisamy 2018), (Feng and Zhang 2020), (Narvaez et al. 2020), (Pang, Niu, and O'Neill 2020), (Zang et al. 2020), (Gao et al. 2020), (Kumari and Toshniwal 2021) **Others –** (Sharma and Kakkar 2018), (Kisi, Heddam, and Yaseen 2019), (Munkhammar, Van Der Meer, and Widen 2019), (Wu et al. 2019a), (Wu et al. 2019b).

3.2.2. Plane of array irradiance estimation

Decomposition models are used to find the three components (global horizontal irradiation, diffuse and direct irradiation) of solar irradiation on a horizontal plane by utilising any component. Transposition models convert these horizontal components to their equivalent in the plane of solar panels. Papers using combination models for the plane of array irradiance estimation are categorised as follows- (Lave et al. 2015), (Cui et al. 2019), (Tina and Ventura 2011), (Khahro et al. 2015), (Passow, Lee, and Panchula 2017), (Riley et al. 2017), (Oka et al. 2018), (Takilalte, Harrouni, and Yaiche 2020), (Meng, Loonen, and Hensen 2020).

3.2.3. PV performance models

PV performance models consist of PV cell and inverter models. PV cell is the basic unit of the photovoltaic system and plays a significant role in PV power forecasting. Inverter model converts instantaneous DC power from PV cell into AC power. Papers using PV performance models for PV power forecasting are categorised as follows- (Cui et al. 2019), (Liaoliao et al. 2017), (Tan and Deng 2017), (Durrani et al. 2018), (Ben Ammar, Ben Ammar, and Oualha 2020).

The summary of literature using the above three categorised models (solar radiation forecasting, plane of array irradiance estimation, PV performance models) for PV power forecasting is given in (Table 2). Readers can refer to these references for more detail.

As discussed in section 1.1, the physical or empirical model is one of the categories of PV forecasting. The empirical models depend upon the physical assumptions of atmosphere, mathematical equations, solar elevation, hour angle, solar declination angle, etc.; this increases the complexity of the model and makes it more suitable for medium and long-

term forecasting applications. In contrast, artificial intelligence applies a learning process to solve the non-linear and complex relationship between the dependent and independent variables (input and output data). Many articles reflect the suitability of machine learning and artificial intelligence techniques over physical models for short-term forecasting. Recently, Ben Ammar, Ben Ammar, and Oualha (2020) proposed three methodologies, including empirical models, artificial intelligence technique and the hybrid model based on a neuro-fuzzy inference system for day-ahead solar irradiance and ambient temperature forecasting. The lowest normalised mean bias error (NMBE) and normalised RMSE values were obtained using the hybrid model combining fuzzy logic and neural networks. Recent studies also show the potential of combining physical models with machine learning and intelligence techniques. An analysis based on this idea has been proposed by Huertas-Tato et al. (2020). This study presents a vision to integrate the results from four forecasting models (satellite-based model, Smart Persistence, WRF-solar and CIADcast) with the popular ML algorithm (SVM) by three blending strategies (general, by horizons and regional). Further, a similar analysis has been performed in which authors compared the performance of several ML models (ELM, SVM, multi-layer perceptron (MLP), and Gaussian process (GP)) with and without considering the various physical models (CAMS, Heliostat-2 and SolarGIS) as input variables (Cornejo-Bueno et al. 2019). The results demonstrated that the ELM exhibits the lowest forecast error compared to other regressors while including the physical models as input.

The researchers are adopting the various architectures of neural networks in solar irradiance forecasting due to their ability to model the nonlinearity and complexity of the data, as discussed in section 3.1. Recently, H. Wang et al. (2020) proposed a four layers-based (one input layer, two linear and one nonlinear layer) direct explainable neural network for 1-min ahead solar irradiance forecasting. The authors concluded that the proposed deep learning model provides better performance accuracy than the traditional neural network, SVM and XGBF, along with higher training efficiency. Further, Feng and Zhang (2020) applied another deep learning architecture, i.e. deep convolutional neural network (CNN), for intra-hour solar irradiance forecasting. The authors developed this model by considering only sky images without any extra feature and reported a good range of normalised RMSE (8.85%) and forecasting skill score (25.14%).

Similar to the literature review on intelligent optimisation algorithms for direct PV power forecasting (section 3.1), some recent articles based on these algorithms for the hyperparameters tuning of ML models have also been reviewed in solar irradiance forecasting. For example, Fan et al. (2020) proposed a model to predict daily diffuse solar irradiance by optimising SVM parameters with three intelligent meta-heuristic approaches (PSO, BAT, and WOA). Along with the recommendation of BAT algorithm, the study also reported an interesting finding of improvement in the prediction results by involving the air pollution parameters. Similarly, Marzouq et al. (2020) applied the crossover and mutation-based evolutionary algorithm, and Y. Feng et al. (2020) involved PSO to optimise the parameters of neural networks.

It is well known that the stochastic nature of PV data is the result of variability in solar irradiance, ambient temperature, and other meteorological variables. This necessitates some data-preprocessing tools to deal with the non-linear and non-stationary data by suitably extracting the embedded information. It can be concluded from the previous literature that wavelet transformation is the most commonly used signal decomposition method. However, it is reported that the wavelet transformation-based ML model achieved better performance than the single-stage models, some drawbacks have been associated with them. Selection of the mother wavelet basis function is the prerequisite of the said data-preprocessing tool. A new multi-resolution time-series decomposition tool (empirical mode decomposition (EMD)) proposed by Huang et al. (1998) may overcome the limitations of wavelet transformation. Recently, Gao et al. (2020) applied a hybrid deep learning model which combines two popular deep neural networks to extract the temporal-spatial information from the CEEMDAN-decomposed solar irradiance subseries. CEEMDAN, the empirical mode decomposition variant, can divide the noisy and volatile time series into several subseries termed intrinsic mode functions. Similarly, Gupta and Singh (2021) proposed a univariate model for 1–11 h ahead GHI forecasting by utilising the EMD and EEMD time-series decomposition techniques. The authors reported a tradeoff between these two techniques in terms of time complexity and performance accuracy. Further, Prasad et al. (2020) proposed a multivariate empirical mode decomposition (MEMD) and singular vector decomposition-based machine learning model for the week ahead solar irradiance forecasting. Unlike EMD and its variants, Multivariate empirical mode decomposition can be applied in the multivariate setting.

After reviewing the vast literature on machine learning based-models for PV forecasting, it can be concluded that no model is universal. Different models experience different limitations such as model complexity in terms of time and architecture, computational cost, instability issues, low performance for highly noisy and complex data, etc. Now-a-days, researchers are interested in designing ensemble or hybrid models to combine the rough prediction of several base learners. Several recent articles on ensemble and hybrid models and their superiority over single-stage models reflect this area's popularity and ongoing research. For example, recently, Zang et al. (2020) proposed a deep learning hybrid model based on CNN-LSTM to model the spatiotemporal correlations between the neighbouring and the target site to predict 1-h ahead global horizontal irradiance. The authors combined the spatial and temporal features extraction ability of CNN and LSTM, respectively, and reported improved performance compared to single-stage models. Similarly, Gao et al. (2020) used the same idea by adding an extra step of the data decomposition technique for the week ahead solar irradiance forecasting.

Further, Lee et al. (2020) presented a comparative analysis of various tree-based ensemble models (bagged trees, boosted trees, random forest and generalised random forest) with the individual regressors (SVM and Gaussian process regressor). The results showed in the article reflect the superiority of ensemble models over single-stage models for short-term solar irradiance forecasting. Later, Kumari and Toshniwal (2021) proposed a novel hybrid model based on a tree-based ensemble and deep learning approach for an hour ahead solar irradiance forecasting. By considering reliable information extraction from the noisy raw data and ML model parameters extraction, Prasad et al. (2019) proposed a multi-stage model. The authors combined the MEMD and ant colony optimisation (ACO) in the first stage, and then the monthly solar irradiance forecasting was performed using the random forest ensemble model.

As discussed earlier, in the indirect way of PV power forecasting, once we have the solar irradiance data on a horizontal plane, it needs to be converted into a corresponding value on a tilted surface. Many studies in this field have introduced various decomposition and transposition models based on mathematical equations to determine the solar irradiance in the plane of PV array from the horizontal one. Recently, Takilalte, Harrouni, and Yaiche (2020) adopted the two popular conventional models Perrin Brichambaut and Liu and Jordan, to convert the 5-min step global irradiance in the horizontal plane into the plane of PV array. The authors reported improved model accuracy by considering the cloudiness factors (direct and diffuse) in the conventional models. Similarly, Meng, Loonen, and Hensen (2020) applied a Perez transposition model to convert the hourly global horizontal solar irradiance into their corresponding values in the tilted plane. The authors also presented an interesting finding in this field by determining the set of tilt and azimuth angles to check the effect of these angles on the profile of plane of array solar irradiance. Recently, Cui et al. (2019) presented an article on indirect PV power forecasting by considering the effect of the air mass, soiling factor, and solar radiation incidence angle. It was concluded from the results that the modified PV power forecasting model shows improved performance accuracy for different weather conditions than the traditional model. The authors also reported the suitability of this model for not only newly developed PV plant but also for the old plant in case of failure of real-time data acquisition.

4. Analysis of the development of literature on PV power forecasting

This section presents the analysis of the development of literature on PV power forecasting based on the following factors: (1) chronological growth of literature, (2) methodology used for forecasting, (3) evolution of techniques with time, (4) forecast horizon and (5) the input parameters for data-driven models. The details are given in the following subsections.

4.1. Growth of literature with time

(Figure 4) shows the chronological development of the literature reviewed in this paper. The first category is PV power direct forecasting, and the remaining three categories, i.e. solar radiation forecasting, the plane of array irradiance estimation and PV performance models, are the sub-categories of PV power indirect forecasting.

4.2. Literature classification based on methods deployed

Data-preparation and dimensionality reduction methods and unsupervised learning-based algorithms cannot be directly applied for the forecasting applications, unlike supervised learning. However, the application of these techniques with forecasting models may increase the accuracy. (Figure 5) shows the literature reviewed in this paper utilising data-driven techniques in the field of PV power forecasting. In (Figure 5), there may be some references that are common in more than one data-driven method.

4.3. Evolution of techniques with time

Increasing the accuracy of PV power forecasting is a challenging task for researchers. A lot of research is going on in this field. (Figure 6) shows the evolution of some ML techniques in this domain with time. It can be seen that deep learning, which is a newly developed sub-branch of machine learning, is being used in the field of PV power forecasting, and its popularity is increasing with time. Ensemble of predictors combines the features of two or more models. As seen in most of the cases in Section 3, its better forecasting accuracy and low error is attracting the researcher's attention. Single-stage models like ANN and SVM are also some of the most commonly used machine learning models because of the quality to deal with the non-linearity of the forecasting problem. In (Figure 6), there may be some references that are common in more than one technique.

4.4. Forecast horizon

Forecast horizon defines the span of time in the future for which the forecasts are to be made. For example 1-day ahead forecast horizon for a specific site and testing period means the forecasts will be made up to one day in the future for the daylight hours (Chaturvedi 2016).

Based on the span of time, forecasting is classified into four different types, and each forecast horizon has its own application.

Very short-term forecast: The forecast for few seconds to several minutes ahead comes in the very short-term forecast category.

Short-term forecast: The forecast for 1-h or several hours to 24 h or 1-week ahead comes in the short-term forecast category.

Medium-term forecast: The forecast from 1 month to 1 year ahead falls in the medium-term forecast category.

Table 2. Summary of literature review on indirect PV power forecasting.

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
Solar radiation forecasting					
(Zeng and Qiao 2013) **USA**	1 h, 2 h, 3 h	1991–2005	Atmospheric transmissivity in 2D, humidity, wind speed and sky cover.	Least-square support vector machine, RBFNN	• A linear model is worse than a non-linear in terms of capturing the non-linear characteristics. • Sky-cover is found important site-independent input variable.
(Shamshirband et al. 2015) **Iran**	1-d	1995–2004	GHI, sunshine hour, maximum, minimum, and average air temperatures and relative humidity	Extreme learning machine (ELM)	• ELM is a highly accurate and reliable model as compared to other ML models.
(Azimi, Ghayekhloo, and Ghofrani 2016) **USA**	1-h, 2-h . . . .48 h	NA	Solar irradiance time-series data	Transformation based (TB) K-means clustering, MLP neural network	• The main feature of TB K-means clustering is to provide consistent results for different runs.
(Belaid and Mellit 2016) **Algeria**	daily and monthly	2 years	Maximum, minimum, mean, maximum-minimum temperatures, horizontal daily global solar radiation	SVM	• The proposed model requires few parameters and low computational time to obtain good accuracy compared to other comparison neural networks.
(Çelik, Teke, and Yildirim 2016) **Turkey**	1-month	10 years	Longitude, latitude, month, temp., sunshine hours, and altitude	ANN with LM algorithm	• ANN model accurately predict the GSR with the use of limited meteorological variables
(Deo, Wen, and Qi 2016) **Australia**	Daily and monthly	1 year 4 months	Sunshine hours, maximum temperature minimum temperature, wind speed, evaporation and precipitation.	SVM-WT	• SVM-WT model found accurate for both daily and monthly forecast horizons.
(Jang et al. 2016) **South Korea**	15 to 300 min	4 years satellite data	Satellite images for atmospheric motion vector (AMV), irradiance images, and cloud analysis images	support vector machine (SVM)	• The proposed model shows better prediction accuracy compared to conventional time series and ANN models.
(J. Li et al. 2016) **Australia**	5 to 30 min	2 months	Weather data (temperature, humidity, atmospheric pressure) and solar data.	Hidden Markov Model (HMM) and SVM regression	• It is observed that as the forecast horizon increases forecast accuracy of HMM and SVM regression models decreases.
(Lou et al. 2016) **China**	1-month	5 years	Global, diffuse and direct normal irradiance, clearness index, air temp, cloud cover, visibility, and solar altitude	Boosted regression tree, logistic regression for multiple variables	• Cloud cover and clearness index are identified as the most important input variables to predict diffuse solar irradiance.
(Jiménez-Pérez and Mora-López 2016) **Spain**	1-d	3 years	Humidity, atmospheric pressure, GHI, and air temperature	k-means clustering, decision trees, ANN, SVM regression (SVM-R), and SVM classification (SVM-C)	• SVM- C and SVM-R combination outperforms the other four classification and regression models combination.
(Aybar-Ruiz et al. 2016) **Spain**	NA	NA	Predictive variables from WRF, previous solar radiation	GGA–ELM	• The outputs from a numerical weather model (WRF) are applied as input features for the ELM.
(Shamshirband et al. 2016) **Malaysia**	1-d	2006–2012	Diffuse irradiance and GHI	SVM-WT	• The authors considered the correlation of cloudiness index with clearness index only as an input variable to predict diffuse solar irradiance.
(H. Sun et al. 2016) **China**	1-d	2001–2012	Sunshine hours, max and min air temperature, air temperature difference, solar radiation, API	Random forest	• RF models with air pollution index (API) performed better than non-API RF and empirical models.
(Gutierrez-Corea et al. 2016) **Spain**	1–6 h	1 year	GHI, air temp, humidity, extraterrestrial solar radiation, wind direction, clearness index, zenith, wind speed	MLP	• For short-term forecast horizon up to 3 h, nearest neighbour data estimates play an important role.
(Wang et al. 2016) **China**	1-d	1961–2014	Sunshine durations, humidity, air temperature, GHI, max and min air temperature, water vapour pressure	MLP, GRNN, RBNN, and improved Bristow-Campbell (IBC)	• The MLP and RBNN models outperform the other comparable models.
(Baser and Demirhan 2017) **Turkey**	24-h	2000 and 2013	Solar radiation, mean air temperature, relative humidity, longitude, latitude, altitude	Fuzzy regression with SVM (FR-SVM)	• Linear, polynomial, Gaussian, and sigmoid kernel functions are used • Gaussian kernel-based FR-SVM approach found better.
(Bouzgou and Gueymard 2017) **Algeria**	15-min, 1-h, 24-h	NA	GHI components	Minimum redundancy-maximum relevance (MRMR), ELM	• The proposed model combines two blocks: the first is dimensionality reduction methodology, and the other is ELM to forecast solar irradiance.
(Hassan et al. 2017) **Egypt**	1 h-24 h	NA	Horizontal global and diffuse irradiance, normal irradiance	Gradient boosting, bagging, and RF	• Bagging and RF ensemble techniques outperformed the gradient boosting methodology.

(*Continued*)

Table 2. (Continued).

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
Solar radiation forecasting					
(Ibrahim and Khatib 2017) **Malaysia**	1-d	1 year	Sunshine ratio, ambient temp, month and day no., hours per day, and humidity	RF-FA	• Firefly algorithm optimises the number of trees and leaves per tree for the RF forecasting model.
(Mousavi, Mostafavi, and Jiao 2017) **Iran**	1-d	1995–2014	Average, max and min of air temp, wind speed, pressure, and relative humidity	ANN-SA (SA-simulated annealing)	• ANN-SA model outperforms the traditional ANN and SVM models to predict GHI.
(Voyant et al. 2017a) **France**	NA	1 year	Measured hourly horizontal global radiation	MLP	• The authors discussed the effect of four kinds of uncertainties on forecasted output
(Alfadda, Rahman, and Pipattanasomporn 2018) **USA**	1-h	1 year 9 months	Ground measured GHI, DHI, and DNI, the hour of the day, the month of the year, zenith, aerosol index data, Angstrom Exponent α, Wind speed and direction,	MLP, SVR, kNN and decision tree	• It is found that DNI is 3–4 times more sensitive to aerosol than GHI.
(Fan et al. 2018) **China**	1-d	2 years	Daily meteorological (sunshine duration and air temperature) and air pollution data (PM2.5, PM10, SO2, NO2, CO, O3, AQI)	Support Vector Machine (SVM)	• PM2.5, PM10, O3, and AQI found more relevant air pollution parameters.
(Fouilloy et al. 2018) **France**	1–6 h	Odeillo 2 years Ajaccio 2 years Tilos 1 year	GHI time series	ARMA, MLP, GP, SVR, Standard and pruned regression trees, Boosted and bagged regression trees, Random forests	• ARMA or MLP found efficient models for low variability location, ARMA or bagged regression for medium variability location, and bagged regression or random forest for high variability location
(Khosravi et al. 2018) **Brazil**	1-h	6 years	Temp, pressure, wind speed, humidity, past time-series solar radiation	Adaptive neuro FIS, MLFFNN, RBFNN, SVR, FIS	• MLFNN and SVR outperformed for exogenous forecasting. • MLFNN, SVR, and adaptive neuro FIS outperformed for endogenous forecasting.
(Meenal and Selvakumar 2018) **India**	1-month	NA	Month, longitude, sunshine duration, humidity, max, and min temperature, latitude	Empirical models, SVM and ANN	• Sunshine duration, max temp, month, and latitude are the most influencing parameters according to WEKA.
(Mishra and Palanisamy 2018) **USA**	1 h, 2 h, 3 h, 4 h	2010–2011	Solar irradiance, air temp, humidity, wind speed, direction and station pressure	Recurrent neural network	• The proposed deep learning forecasting model outperforms other ML models in terms of forecasting accuracy.
(Pedro et al. 2018) **USA**	5–30 min	1 year	GHI, DNI and sky images	Point forecast (persistence, KNN, GB, genetic boosting) and probabilistic forecast (persistence ensemble, KNN and GB)	• The authors performed deterministic and probabilistic forecast of GHI and DNI using various ML techniques.
(Sharma and Kakkar 2018) **India**	1 h-48 h	5 years	Solar radiation	FoBa, Leap Forward, spike slab, Cubist, bag Earth GCV	• Leap Forward, FoBa, and Cubist models performed well for large datasets, spikslab with less dataset and bag Earth GCV found independent with a dataset.
(Rao, Premalatha, and Naveen 2018) **India**	1-d	3 years	Max-min temperature, DT, approximate sunshine hours(So), terrestrial radiation(Ho), theoretical sunshine hours(Sa)	ANN	• ANN models outperformed the empirical models.
(S. Sun et al. 2018) **China**	1, 3, and 6 step ahead	8 years 5 months	Solar irradiance	Decomposition-clustering-ensemble (DCE) learning (EEMD-LSSVR-K-LSSVR)	• The DCE learning approach employed EEMD for decomposition, K-means for clustering analysis, and LSSVR for ensemble learning.
(Yao et al. 2018) **China**	1-d	3 years 3 months	GHI, sunshine hours, temp, humidity and AQI	SVM (SVM-1, SVM-2)	• Air quality index parameter improved the results of GHI prediction accuracy.
(Marchesoni-Acland et al. 2019) **France**	10–240 min	2 years	GHI, satellite albedo, local short-term variability	ARMAX	• PGM-S (past ground measurements with satellite data) model outperformed.
(Benali et al. 2019) **Algeria**	1–6 h	3 years	Global, beam and diffuse radiation data	Smart persistence, ANN and RF	• Normal and diffuse horizontal irradiance showed more sensitivity to meteorological variables than global irradiance.
(Cornejo-Bueno et al. 2019) **Spain**	NA	1 year	Cloud index, clear sky solar radiation model, several reflectivity	SVR, extreme learning machine(ELM), GP, MLP	• ELM was found as the best performing regressor among all the machine learning regressors and physical models.

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Table 2. (Continued).

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
Solar radiation forecasting					
(Kisi, Heddam, and Yaseen 2019) **Turkey**	1-month	1983–2010 and 1968–2015 for stations	Solar irradiance, air temperature	Dynamic evolving neural-FIS model (DENFIS)	• DENFIS model provided accurate SR prediction results as compare to MARS, M5Tree, and LSSVR models.
(Lan et al. 2019) **China**	1-d	1 year	Time-series irradiance data	Discrete Fourier transform (DFT), PCA, Elman based neural network	• The proposed model extracts the frequency features, unlike traditional feature extraction in the time domain
(Marzouq et al. 2019) **Morocco**	1-d	6 years	GHI, humidity, rainfall, air temperature, wind direction and speed	Evolutionary artificial neural networks	• The proposed method used genetic operators to predict the solar radiation and automatically find the most affecting input parameters.
(Munkhammar, Van Der Meer, and Widen 2019) **Sweden**	1-d	1 year	GHI, clear sky irradiance	Markov-chain mixture (MCM) distribution and copula model	• MCM model outperforms the copula model and shows nearly the same results with benchmark quantile regression probabilistic model.
(Prasad et al. 2019) **Australia**	1 month	1905–2018	T_{max}, T_{min}, Rain, vapour pressure, max and min relative humidity, Potential Evapotranspiration (FAO5).	MEMD-ACO-RF	• The proposed model outperformed the other comparable models in terms of good performance metrices and low forecast error.
(Wu et al. 2019a) **USA**	24-hour	May to September 2014–2017	Lifting condensation level, inversion strength tropospheric total water mixing ratio, water potential temperature, surface dew point temperature and precipitable water	Analog ensemble	• AnEn outperformed persistence and Weather research and forecasting model in terms of low centred RMSE and RMSE values.
(Wu et al. 2019b) **China**	1-d	2001–2015	GHI, sunshine duration, humidity, max and min temperature, and precipitation	KNEA and CatBoost,	• The proposed two models accurately forecast the GHI in the humid region.
(Yagli, Yang, and Srinivasan 2019) **USA**	1-h	4 years	Solar irradiance	68 machine learning models	• The authors performed an analysis to determine which model is more successful under which climate.
(Fan et al. 2020) **China**	1-d	3 years	GHI, diffuse solar irradiance, sunshine duration, max and min temp, PM2.5, PM10 and O3	SVM-PSO, SVM-BAT and SVM-WOA	• PSO, BAT, and WOA are heuristic algorithms. • SVM-BAT enhanced performance accuracy followed by SVM-PSO and SVM-WOA
(Y. Feng et al. 2020) **China**	1-d	1961–2016	Solar irradiance, sunshine hours, relative humidity, average, max and min air temp.	PSO-ELM	• PSO improves the forecasting accuracy by optimising parameters of ELM.
(Prasad et al. 2020) **Australia**	1-week	1905–2018	Potential evapotranspiration (FAO56), vapour pressure, evaporation, relative humidity at max and min temp., rainfall, max and min temp.	MEMD-SVD- RF	• The authors designed a MEMD-SVD based machine learning model for reducing the feature dimension.
(Yagli et al. 2020) **Singapore**	1-h	3 years	Ground and satellite-based solar irradiance	Cubist, SVR, GLMNET, Random Forest, PPR	• Satellite-based measurements can be taken as an alternate option in solar resource forecasting due to the scarcity of reliable Ground-based data sources.
(Babar et al. 2020) **Norway**	1-d	_	GHI, clear sky indices, solar zenith	Random forest and CLARA-A2, ERA5 satellite-based models	• The random forest has been found as a good alternative to both ERA5 and CLARA-A2 reanalyses and satellite models.
(C. Feng and Zhang 2020) **USA**	10 min to 60 min ahead	6 years	GHI, DNI, DHI, temperature, relative humidity, wind speed, atmosphere pressure	Deep convolutional neural network(CNN)	• CNN has been developed for univariate data (GHI only). • Other variables are used in benchmark models.
(Narvaez et al. 2020) **Colombia**	Daily (24 h, 72 h, 168 h), weekly (168 h, 336 h, 672 h)	2006–2017	GHI, DHI, DNI, solar zenith, temperature, wind speed	Machine learning (Random forest, quantile mapping), deep learning (LSTM, GRU)	• The authors proposed a methodology on site adaption technique (selection of best of both satellite and in-situ measurements) using ML algorithms.
(Marzouq et al. 2020) **Morocco**	1–6 h ahead	1 year	Solar irradiance	HAEANN (History and architecture evolutionary ANN)	• For all three scenarios, the proposed model outperformed the smart persistence, RF, and decision tree models.

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Table 2. (Continued).

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
Solar radiation forecasting					
(Pang, Niu, and O'Neill 2020) **USA**	10 min, 30 min, 60 min	May 22nd to 29 May 2016	Solar irradiance, dry-bulb air temperature, time of day	ANN, ANN_Moving window, RNN, RNN_Moving window	• Moving window algorithm can slightly improve the results. • RNN model outperformed the ANN, but it requires large computational time.
(Ben Ammar, Ben Ammar, and Oualha 2020) **Tunisia**	1-d	_	Temperature, solar irradiance	Empirical models, FFNN, ANFIS (Adaptive neuro-FIS)	• By utilising forecasted (ANFIS outperforms) solar irradiance and temperature, PV power has been calculated by applying a mathematical model.
(Zambrano and Giraldo 2020) **Colombia**	1-h	2005–2017	GHI	ANN, Algorithms for the selection of sites (Mahalanobis distance, kernel machines, PCA, SFS)	• The paper presents some algorithms for selecting similar pattern sites for the prediction of solar irradiance without on-site training measurements.
(Zang et al. 2020) **China**	1-h	2006–2012	GHI, wind speed, and direction, dew point temp, precipitable water, solar zenith, relative humidity, temperature	CNN-LSTM	• The proposed model combines the spatiotemporal correlation of data to predict the GHI effectively.
(Gao et al. 2020) **China**	1-h	6 years	Solar irradiance	CEEMDAN-CNN-LSTM	• Paper presents an advanced technique to decompose the highly noisy solar irradiance time series into different IMFs.
(H. Wang et al. 2020) **China**	1-min	2018	Solar radiation, relative humidity, sun altitude, temperature, wind speed, global and zenith illuminance, etc.	Direct explainable neural network (DXNN)	• Straightforward, and understandable input-output mapping of the proposed prediction model is the critical finding of this research work.
(Lee et al. 2020) **Saudi Arabia**	1-h	_	Solar radiation, month, day of month and hour of the day	Boosted trees, bagged trees, Random forest, generalised random forest	• Ensemble learning models outperformed SVM and Gaussian process regression for 16 evaluations out of 18.
(Huertas-Tato et al. 2020) **Spain**	15 min to 6 h	2 years	GHI, DHI, DNI	Satellite model, WRF-solar, smart persistence, CIADcast, SVM	• Blended physical model using SVM outperforms physical models.
(Kumari and Toshniwal 2021) **India**	1-h	10 years	GHI, relative humidity, temperature, wind speed and direction, pressure, clear sky index	XGBF-DNN(Extreme gradient boosting- Deep neural network)	• The proposed model has reduced the forecasting error by 40% compared to other traditional ML models.
(Gupta and Singh 2021) **India**	1–11 h	5 years	Time series of GHI	EEMD-RF, EMD-RF, EEMD-SVM, EMD-SVM	• A univariate model has been presented for 1–11 h ahead GHI forecasting comparing the two adaptive time-series decomposition techniques.
Plane of array irradiance estimation					
(Tina and Ventura 2011) **Italy**	NA	NA	Global solar radiation for horizontal and 45° and 60° tilted surface	Perez, Klucher, ANN	• Perez and Klucher models are useful to compute POA irradiance when horizontal solar radiations are available.
(Khahro et al. 2015) **Pakistan**	NA	NA	Sunshine hours, temperature, relative humidity, pressure, global solar radiation	Diffuse solar irradiance model	• Tilt angle adjustment two times in a year was found advantageous.
(Lave et al. 2015) **USA**	NA	NA	GHI, DHI, POA, clearness index, solar elevation angle	12 decomposition and 4 transposition models	• Erbs and DIRINT decomposition models showed better performance. • Hay/Davis model outperformed in transposition models.
(Passow, Lee, and Panchula 2017) **USA**	NA	NA	Measured GHI	Decomposition models (DIRINT, Erbs, Reindl-3)	• DIRINT model gave better results, followed by Erbs and Reindl-3 model.
(Riley et al. 2017) **California**	NA	NA	GHI, DHI, POAI	Transposition models(Hays/Davis, Perez)	• By tuning satellite-derived GHI with ground-based GHI, a reduction in error of modelled POAI (evaluated by transposition models) can be observed.
(Oka et al. 2018) **Japan**	NA	NA	Global horizontal irradiance	Four decomposition and 3 transposition models	• Erbs and Perez found the most suitable decomposition and transposition models, respectively.

(*Continued*)

Table 2. (Continued).

Ref., location	Forecast horizon	Data duration	Input variables	Methodology	Findings
Solar radiation forecasting					
(Takilalte, Harrouni, and Yaiche 2020) **Algeria**	5-min	NA	GHI	Perrin Brichambaut, Liu-Jordan models in conjunction with cloudiness factor	• By taking advantage of the cloudiness factor, the proposed model outperformed traditional models and recently developed techniques like ANN.
(Meng, Loonen, and Hensen 2020) **Netherland**	1-h	NA	PV generation and off-site solar irradiance	Perez transposition model	• The authors present a novel technique to find the tilt and azimuth angles for any distributed photovoltaic systems by utilising the Perez transposition model.
PV performance models					
(Liaoliao et al. 2017) **China**	NA	NA	Solar irradiance, atmospheric pressure, ambient temp., wind speed	Three decomposition, 3 transposition models, and PV performance model	• Out of the 9 models, the model with Liu-Jordan performed better as compared to other models.
(Tan 2017) **China**	ultra short term	1 year	Cloud cover, solar irradiance	Adaboost, KNN clustering, and Markov Chain, hotel model, PV engineering model	• The authors introduced the concept of the attenuation coefficient of solar radiation to improve the efficiency of forecasted power
(Durrani et al. 2018) **United States**	1-d ahead	2010–2015	Temperature, pressure, humidity, wind speed and direction, sunshine hours and cloud cover	Combination model, Feed forward neural network, photovoltaic model	• The GHI predicted from MFFNN outperforms the persistence model. • PV model determines PV power through a mathematical equation.
(Cui et al. 2019) **China**	15 min	NA	Solar irradiance, ambient temperature, wind speed, atmospheric pressure,	Three decomposition and four transposition models, PV electric model	• Air mass and soiling effect improved the forecasting results • Erbs+Liu-Jordan model was found better.

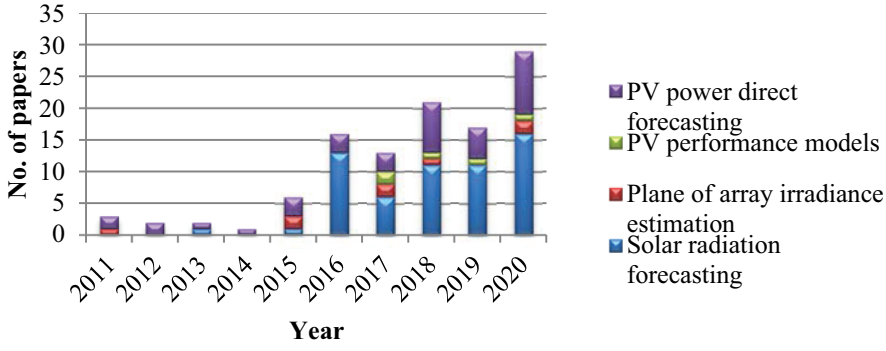


Figure 4. Growth of literature with time.

Long-term forecast: The forecast from 1-year to several years ahead comes in the long-term forecast category.

(Figure 7) shows the classification of reviewed literature for various forecast horizons. For very short-term and short-term forecast horizon, the researchers recommend machine learning models. In (Figure 7 and 8), there may be some references that are common in more than one forecast horizon classification.

4.5. Input parameters

In solar energy research, various parameters are essential. Solar irradiance is the most important parameter in this area of study. The variability of solar irradiance directly affects the power generation of the PV plant. Several meteorological variables, for example temperature, wind direction and speed, sunshine duration, humidity and clou: cover, are also used to estimate solar power generation. The correlation factor of several weather variables with PV power varies, depending upon the location of the PV plant. Based on the nature of input parameters, PV power forecasting can be classified as follows.

Historical data-based forecasting: This type of forecasting identifies a relationship between the input and the output data. Historical solar radiation data/PV power is used as the input to predict next-day solar radiation/PV power.

Historical and meteorological data-based forecasting- This type of forecasting combines historical solar irradiation/PV power with meteorological input parameters like wind direction and speed, humidity, air temperature, cloud cover, aerosol index, air pollution, sky images, etc. to predict solar irradiation/PV power.

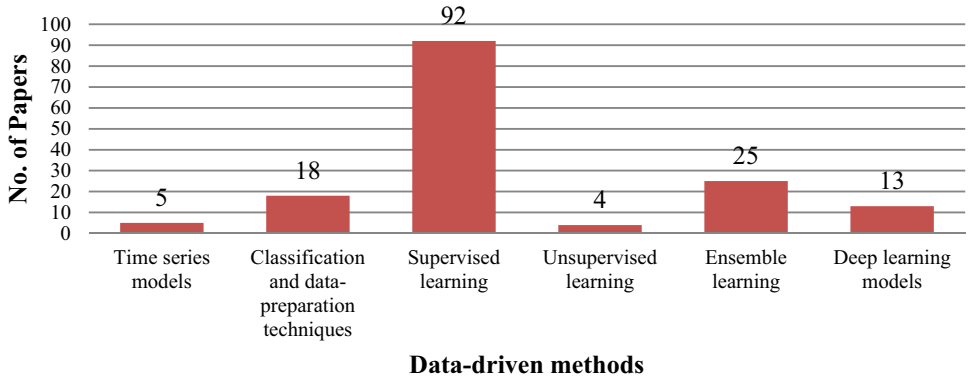


Figure 5. Classification of data-driven methods-based literature.

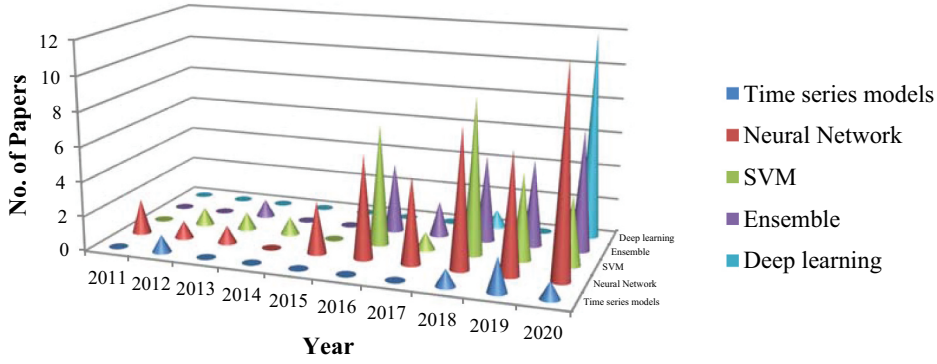


Figure 6. Evolution of techniques with time.

5. Conclusion

The paper presents a review of the recently published papers on PV power forecasting along two pathways, i.e. direct forecasting and indirect forecasting. Direct forecasting techniques directly forecast the PV power output. The literature for indirect PV power forecasting is classified into three categories, i.e. solar radiation forecasting, plane of array irradiance estimation and PV performance models. These three categories are the three steps of the methodology for forecasting PV power in an indirect way. Analysis of the development of literature on PV power forecasting based on some factors like chronological growth of literature, methods deployed, the evolution of techniques, forecast horizon and input parameters has also been presented. Some key points from the literature are as follows.

(1) Researchers suggest an indirect way of PV power forecasting when direct PV power forecasting is not possible using ML techniques. For example, for a newly developed PV plant.
(2) As shown in the paper, many data-driven techniques are available for forecasting. Some are beginning to be used more frequently (bagging, boosting, decision trees, etc.), some are often used (ANN, SVM, ARIMA, MLR), and some have not been used so frequently (deep learning).

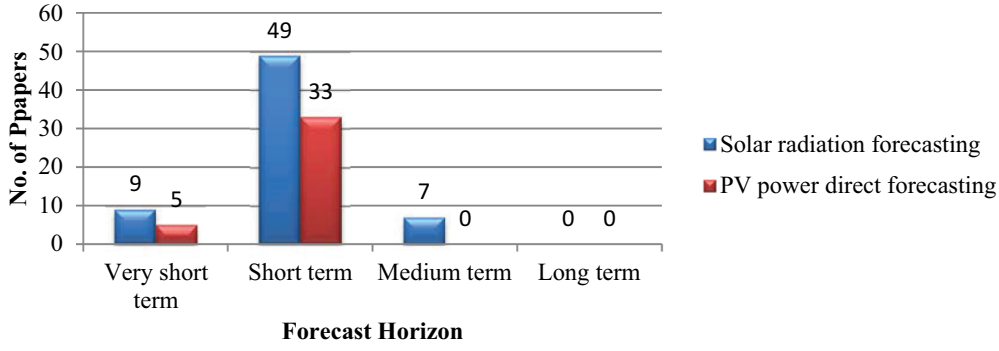


Figure 7. Classification with respect to forecast horizon.

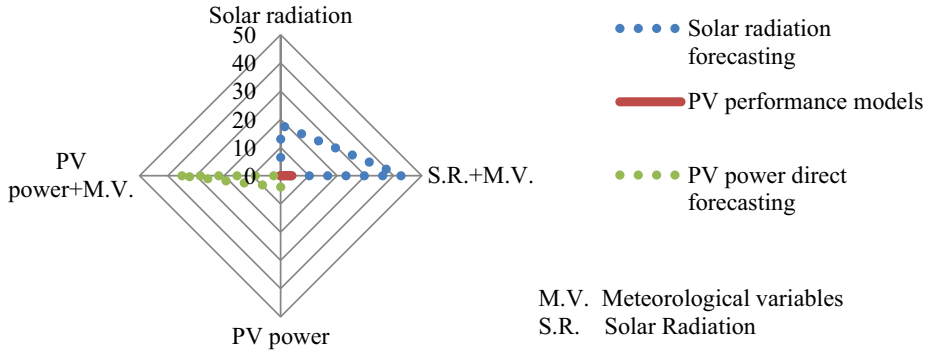


Figure 8. Classification with respect to input parameters.

(3) Ensemble learning-based forecasting models outperform the single-stage forecasting models (ANN, SVM, k-NN, etc.) in most of the cases.
(4) Deep learning is becoming a more powerful tool in time series forecasting problems due to its ability to discover the complex non-linearity of the time series data. The ability of deep learning models to overcome the limitations of traditional time series models (ARMA, ARIMA) makes them the best option for time series forecasting.
(5) From the literature on the plane of array irradiance estimation models, it can be said that, decomposition models' results (modelled and measured) vary with location, whereas transposition models' results vary more with the type of model.

To conclude, we cannot label any model as a universal model. One model can be superior to others in some cases, whereas some other models might perform better in other conditions.

Highlights

1. A review of PV power forecasting using data-driven models has been done.
2.Two pathways for forecasting have been reviewed: direct and indirect.
3. Direct forecasting directly forecasts the PV power output.
4. Indirect forecasting consists of solar irradiance forecast, plane of array irradiance estimation and PV performance model.
5. Analysis of the development of literature in the area has also been presented.

Nomenclature

AnEn	Analog Ensemble
ANN	Artificial neural network
ACO	Ant colony optimisation
ARIMA	Auto regressive integrated moving average
AQI	Air quality index
ARMAX	ARMA with exogenous inputs
BP	Back propagation
BAT	Bat algorithm
BI-LSTM	Bi directional LSTM
BPNN	Back propagation neural network
CatBoost	Gradient boosting with categorical features support
CF	Cascade forward
CLARA-A2	Cloud, Albedo, Radiation Dataset Edition 2
CEEMDAN	Complete ensemble empirical mode decomposition adaptive noise
DNI	Direct normal irradiance
DNN-RODDPSO	Deep neural network- randomly occurring distributed delayed particle swarm optimisation
ECMWF	European centre for medium range weather forecasts
EEMD	Ensemble Empirical Mode Decomposition
ERA5	ECMWF Reanalysis 5
FA	Firefly algorithm
FFNN	Feed forward neural network
FIS	Fuzzy inference system
FoBa	Adaptive forward backward greedy algorithm
GA-ANN	Genetic algorithm-artificial neural network
GGA	Grouping genetic algorithm
GHI	Global horizontal irradiance
GRNN	General regression NN
GLMNET	Generalised linear model with elastic net regulariser
GRU	Gated recurrent
GPR	Gaussian process regression
IBL	Inversion based height
IMF	Intrinsic mode function
K-NN	K-nearest neighbour

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Notes on contributors

Priya Gupta is currently pursuing a Ph.D. in Solar Energy in the Department of Hydro and Renewable Energy at IIT Roorkee, India. She received the Master of Technology degree (Gold Medal) in Power System from NIT Kurukshetra (2019), and a Bachelor of Technology degree in Electrical Engineering from G. B. Pant Engineering College, Pauri Garhwal (2016). Her main research areas are machine learning, deep learning, time-series data, solar resource forecasting, MPPT control of PV modules. She is a recipient of the Ministry of Education, Government of India fellowship for Postgraduate research as well as Doctoral Research.

Rhythm Singh is currently working as an Assistant Professor in the Department of Hydro and Renewable Energy at IIT Roorkee, India. He holds a Ph.D. in Energy Science and Engineering from IIT Bombay (2018), a Master of Technology degree in Energy Studies from IIT Delhi (2009), and a Bachelor of Technology (Gold Medal) degree in Electrical Engineering from MNNIT Allahabad (2007). His research interests lie at the intersection of technological analysis and policy support for enabling a transition to renewable energy technologies. He was awarded the Best Poster Award at the 2017 IEEE 44th Photovoltaic Specialists Conference held in Washington D.C., USA, for his paper on solar resource variability assessment. He is an active reviewer for around 13 reputed journals published by Elsevier, Taylor and Francis, IEEE, IET, and AIP publishing.

ORCID

Rhythm Singh http://orcid.org/0000-0003-4510-5845

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